

XI.

The Role of Modern Physics in the Present Development of Human Thinking

THE philosophical implications of modern physics have been discussed in the foregoing chapters in order to show that this most modern part of science touches very old trends of thought at many points, that it approaches some of the very old problems from a new direction. It is probably true quite generally that in the history of human thinking the most fruitful developments frequently take place at those points where two different lines of thought meet. These lines may have their roots in quite different parts of human culture, in different times or different cultural environments or different religious traditions; hence if they actually meet, that is, if they are at least so much related to each other that a real interaction can take place, then one may hope that new and interesting developments will follow. Atomic physics as a part of modern science does actually penetrate in our time into very different cultural traditions. It is not only taught in Europe and the Western countries, where it belongs to the traditional activity in the natural sciences, but it is also studied in the Far East, in countries like Japan and China and India,

with their quite different cultural background, and in Russia, where a new way of thinking has been established in our time; a new way related both to specific scientific developments of the Europe of the nineteenth century and to other entirely different traditions from Russia itself. It can certainly not be the purpose of the following discussion to make predictions about the probable result of the encounter between the ideas of modern physics and the older traditions. But it may be possible to define the points from which the interaction between the different ideas may begin.

In considering this process of expansion of modern physics it would certainly not be possible to separate it from the general expansion of natural science, of industry and engineering, of medicine, etc., that is, quite generally of modern civilization in all parts of the world. Modern physics is just one link in a long chain of events that started from the work of Bacon, Galileo and Kepler and from the practical application of natural science in the seventeenth and eighteenth centuries. The connection between natural science and technical science has from the beginning been that of mutual assistance: The progress in technical science, the improvement of the tools, the invention of new technical devices have provided the basis for more, and more accurate, empirical knowledge of nature; and the progress in the understanding of nature and finally the mathematical formulation of natural laws have opened the way to new applications of this knowledge in technical science. For instance, the invention of the telescope enabled the astronomers to measure the motion of the stars more accurately than before; thereby a considerable progress in astronomy and in mechanics was made possible. On the other hand, precise knowledge of the mechanical laws was of the greatest value for the improvement of mechanical tools,

for the construction of engines, etc. The great expansion of this combination of natural and technical science started when one had succeeded in putting some of the forces of nature at the disposal of man. The energy stored up in coal, for instance, could then perform some of the work which formerly had to be done by man himself. The industries growing out of these new possibilities could first be considered as a natural continuation and expansion of the older trades; at many points the work of the machines still resembled the older handicraft and the work in the chemical factories could be considered as a continuation of the work in the dyehouses and the pharmacies of the older times. But later entirely new branches of industry developed which had no counterpart in the older trades; for instance, electrical engineering. The penetration of science into the more remote parts of nature enabled the engineers to use forces of nature which in former periods had scarcely been known; and the accurate knowledge of these forces in terms of a mathematical formulation of the laws governing them formed a solid basis for the construction of all kinds of machinery.

The enormous success of this combination of natural and technical science led to a strong preponderance of those nations or states or communities in which this kind of human activity flourished, and as a natural consequence this activity had to be taken up even by those nations which by tradition would not have been inclined toward natural and technical sciences. The modern means of communication and of traffic finally completed this process of expansion of technical civilization. Undoubtedly the process has fundamentally changed the conditions of life on our earth; and whether one approves of it or not, whether one calls it progress or danger, one must realize that it has gone far beyond any control through human forces. One

may rather consider it as a biological process on the largest scale whereby the structures active in the human organism encroach on larger parts of matter and transform it into a state suited for the increasing human population.

Modern physics belongs to the most recent parts of this development, and its unfortunately most visible result, the invention of nuclear weapons, has shown the essence of this development in the sharpest possible light. On the one hand, it has demonstrated most clearly that the changes brought about by the combination of natural and technical sciences cannot be looked at only from the optimistic viewpoint; it has at least partly justified the views of those who had always warned against the dangers of such radical transmutation of our natural conditions of life. On the other hand, it has compelled even those nations or individuals who tried to keep apart from these dangers to pay the strongest attention to the new development, since obviously political power in the sense of military power rests upon the possession of atomic weapons. It can certainly not be the task of this volume to discuss extensively the political implications of nuclear physics. But at least a few words may be said about these problems because they always come first into the minds of people when atomic physics is mentioned.

It is obvious that the invention of the new weapons, especially of the thermonuclear weapons, has fundamentally changed the political structure of the world. Not only has the concept of independent nations or states undergone a decisive change, since any nation which is not in possession of such weapons must depend in some way on those very few nations that do produce these arms in large quantity; but also the attempt of warfare on a large scale by means of such weapons has become practically an absurd kind of suicide. Hence one frequently hears the optimistic

view that therefore war has become obsolete, that it will not happen again. This view, unfortunately, is a much too optimistic oversimplification. On the contrary, the absurdity of warfare by means of thermonuclear weapons may, in a first approximation, act as an incentive for war on a small scale. Any nation or political group which is convinced of its historical or moral right to enforce some change of the present situation will feel that the use of conventional arms for this purpose will not involve any great risks; they will assume that the other side will certainly not have recourse to the nuclear weapons, since the other side being historically and morally wrong in this issue will not take the chance of war on a large scale. This situation would in turn induce the other nations to state that in case of small wars inflicted upon them by aggressors, they would actually have recourse to the nuclear weapons, and thus the danger obviously remains. It may quite well be that in about twenty or thirty years from now the world will have undergone so great changes that the danger of warfare on a large scale, of the application of all technical resources for the annihilation of the opponent, will have greatly diminished or disappeared. But the way to this new state will be full of the greatest dangers. We must as in all former times, realize that what looks historically or morally right to the one side may look wrong to the other side. The continuation of the status quo may not always be the correct solution; it may, on the contrary, be most important to find peaceful means of adjustments to new situations, and it may in many cases be extremely difficult to find any just decision at all. Therefore, it is probably not too pessimistic to say that the great war can be avoided only if all the different political groups are ready to renounce some of their apparently most obvious rights—in view of the fact that the question of right or wrong may look essen-

tially different from the other side. This is certainly not a new point of view; it is in fact only an application of that human attitude which has been taught through many centuries by some of the great religions.

The invention of nuclear weapons has also raised entirely new problems for science and scientists. The political influence of science has become very much stronger than it was before World War II, and this fact has burdened the scientist, especially the atomic physicist, with a double responsibility. He can either take an active part in the administration of the country in connection with the importance of science for the community; then he will eventually have to face the responsibility for decisions of enormous weight which go far beyond the small circle of research and university work to which he was wont. Or he may voluntarily withdraw from any participation in political decisions; then he will still be responsible for wrong decisions which he could possibly have prevented had he not preferred the quiet life of the scientist. Obviously it is the duty of the scientists to inform their governments in detail about the unprecedented destruction that would follow from a war with thermonuclear weapons. Beyond that, scientists are frequently requested to participate in solemn resolutions in favor of world peace; but considering this latter demand I must confess that I have never been able to see any point in declarations of this kind. Such resolutions may seem a welcome proof of goodwill; but anyone who speaks in favor of peace without stating precisely the conditions of this peace must at once be suspected of speaking only about that kind of peace in which he and his group thrive best—which of course would be completely worthless. Any honest declaration for peace must be an enumeration of the sacrifices one is pre-

pared to make for its preservation. But as a rule the scientists have no authority to make statements of this kind.

At the same time the scientist can do his best to promote international co-operation in his own field. The great importance that many governments attach to research in nuclear physics nowadays and the fact that the level of scientific work is still very different in different countries favors international co-operation in this work. Young scientists of many different countries may gather in research institutions in which a strong activity in the field of modern physics is going on and the common work on difficult scientific problems will foster mutual understanding. In one case, that of the Geneva organization, it has even been possible to reach an agreement between a number of different nations for building a common laboratory and for constructing by a combined effort the expensive experimental equipment for research in nuclear physics. This kind of co-operation will certainly help to establish a common attitude toward the problems of science—common even beyond the purely scientific problems—among the younger generation of scientists. Of course one does not know beforehand what will grow out of the seeds that have been sown in this way when the scientists return into their old environments and again take part in their own cultural traditions. But one can scarcely doubt that the exchange of ideas between young scientists of different countries and between the different generations in every country will help to approach without too much tension that new state of affairs in which a balance is reached between the older traditional forces and the inevitable necessities of modern life. It is especially one feature of science which makes it more than anything else suited for establishing the first strong connection between different cultural traditions. This is the fact that the ultimate decisions about the value of a

special scientific work, about what is correct or wrong in the work, do not depend on any human authority. It may sometimes take many years before one knows the solution of a problem, before one can distinguish between truth and error; but finally the questions will be decided, and the decisions are made not by any group of scientists but by nature itself. Therefore, scientific ideas spread among those who are interested in science in an entirely different way from the propagation of political ideas.

While political ideas may gain a convincing influence among great masses of people just because they correspond or seem to correspond to the prevailing interests of the people, scientific ideas will spread only because they are true. There are objective and final criteria assuring the correctness of a scientific statement.

All that has here been said about international co-operation and exchange of ideas would of course be equally true for any part of modern science; it is by no means confined to atomic physics. In this respect modern physics is just one of the many branches of science, and even if its technical applications—the arms and the peaceful use of atomic energy—attach a special weight to this branch, there would be no reason for considering international co-operation in this field as far more important than in any other field. But we have now to discuss again those features of modern physics which are essentially different from the previous development of natural science, and we have for this purpose once more to go back to the European history of this development that was brought about by the combination of natural and technical sciences.

It has frequently been discussed among the historians whether the rise of natural science after the sixteenth century was in any way a natural consequence of earlier trends in human thinking.

It may be argued that certain trends in Christian philosophy led to a very abstract concept of God, that they put God so far above the world that one began to consider the world without at the same time also seeing God in the world. The Cartesian partition may be called a final step in this development. Or one may point out that all the theological controversies of the sixteenth century produced a general discontent about problems that could not really be settled by reason and were exposed to the political struggles of the time; that this discontent favored interest in problems which were entirely separated from the theological disputes. Or one may simply refer to the enormous activity, the new spirit that had come into the European societies through the Renaissance. In any case during this period a new authority appeared which was completely independent of Christian religion or philosophy or of the Church, the authority of experience, of the empirical fact. One may trace this authority back into older philosophical trends, for instance, into the philosophy of Occam and Duns Scotus, but it became a vital force of human activity only from the sixteenth century onward. Galileo did not only *think* about the mechanical motions, the pendulum and the falling stone; he tried out by experiments, quantitatively, how these motions took place. This new activity was in its beginning certainly not meant as a deviation from the traditional Christian religion. On the contrary, one spoke of two kinds of revelation of God. The one was written in the Bible and the other was to be found in the book of nature. The Holy Scripture had been written by man and was therefore subject to error, while nature was the immediate expression of God's intentions.

However, the emphasis on experience was connected with a slow and gradual change in the aspect of reality. While in the Middle Ages what we nowadays call the symbolic meaning of a

thing was in some way its primary reality, the aspect of reality changed toward what we can perceive with our senses. What we can see and touch became primarily real. And this new concept of reality could be connected with a new activity: we can experiment and see how things really are. It was easily seen that this new attitude meant the departure of the human mind into an immense field of new possibilities, and it can be well understood that the Church saw in the new movement the dangers rather than the hopes. The famous trial of Galileo in connection with his views on the Copernican system marked the beginning of a struggle that went on for more than a century. In this controversy the representatives of natural science could argue that experience offers an undisputable truth, that it cannot be left to any human authority to decide about what really happens in nature, and that this decision is made by nature or in this sense by God. The representatives of the traditional religion, on the other hand, could argue that by paying too much attention to the material world, to what we perceive with our senses, we lose the connection with the essential values of human life, with just that part of reality which is beyond the material world. These two arguments do not meet, and therefore the problem could not be settled by any kind of agreement or decision.

In the meantime natural science proceeded to get a clearer and wider picture of the material world. In physics this picture was to be described by means of those concepts which we nowadays call the concepts of classical physics. The world consisted of things in space and time, the things consist of matter, and matter can produce and can be acted upon by forces. The events follow from the interplay between matter and forces; every event is the result and the cause of other events. At the same time the human attitude toward nature changed from a contemplative

one to the pragmatic one. One was not so much interested in nature as it is; one rather asked what one could do with it. Therefore, natural science turned into technical science; every advancement of knowledge was connected with the question as to what practical use could be derived from it. This was true not only in physics; in chemistry and biology the attitude was essentially the same, and the success of the new methods in medicine or in agriculture contributed essentially to the propagation of the new tendencies.

In this way, finally, the nineteenth century developed an extremely rigid frame for natural science which formed not only science but also the general outlook of great masses of people. This frame was supported by the fundamental concepts of classical physics, space, time, matter and causality; the concept of reality applied to the things or events that we could perceive by our senses or that could be observed by means of the refined tools that technical science had provided. Matter was the primary reality. The progress of science was pictured as a crusade of conquest into the material world. Utility was the watchword of the time.

On the other hand, this frame was so narrow and rigid that it was difficult to find a place in it for many concepts of our language that had always belonged to its very substance, for instance, the concepts of mind, of the human soul or of life. Mind could be introduced into the general picture only as a kind of mirror of the material world; and when one studied the properties of this mirror in the science of psychology, the scientists were always tempted—if I may carry the comparison further—to pay more attention to its mechanical than to its optical properties. Even there one tried to apply the concepts of classical physics, primarily that of causality. In the same way life

was to be explained as a physical and chemical process, governed by natural laws, completely determined by causality. Darwin's concept of evolution provided ample evidence for this interpretation. It was especially difficult to find in this framework room for those parts of reality that had been the object of the traditional religion and seemed now more or less only imaginary. Therefore, in those European countries in which one was wont to follow the ideas up to their extreme consequences, an open hostility of science toward religion developed, and even in the other countries there was an increasing tendency toward indifference toward such questions; only the ethical values of the Christian religion were excepted from this trend, at least for the time being. Confidence in the scientific method and in rational thinking replaced all other safeguards of the human mind.

Coming back now to the contributions of modern physics, one may say that the most important change brought about by its results consists in the dissolution of this rigid frame of concepts of the nineteenth century. Of course many attempts had been made before to get away from this rigid frame which seemed obviously too narrow for an understanding of the essential parts of reality. But it had not been possible to see what could be wrong with the fundamental concepts like matter, space, time and causality that had been so extremely successful in the history of science. Only experimental research itself, carried out with all the refined equipment that technical science could offer, and its mathematical interpretation, provided the basis for a critical analysis—or, one may say, enforced the critical analysis—of these concepts, and finally resulted in the dissolution of the rigid frame.

This dissolution took place in two distinct stages. The first was the discovery, through the theory of relativity, that even

such fundamental concepts as space and time could be changed and in fact must be changed on account of new experience. This change did not concern the somewhat vague concepts of space and time in natural language; but it did concern their precise formulation in the scientific language of Newtonian mechanics, which had erroneously been accepted as final. The second stage was the discussion of the concept of matter enforced by the experimental results concerning the atomic structure. The idea of the reality of matter had probably been the strongest part in that rigid frame of concepts of the nineteenth century, and this idea had at least to be modified in connection with the new experience. Again the concepts so far as they belonged to the natural language remained untouched. There was no difficulty in speaking about matter or about facts or about reality when one had to describe the atomic experiments and their results. But the scientific extrapolation of these concepts into the smallest parts of matter could not be done in the simple way suggested by classical physics, though it had erroneously determined the general outlook on the problem of matter.

These new results had first of all to be considered as a serious warning against the somewhat forced application of scientific concepts in domains where they did not belong. The application of the concepts of classical physics, e.g., in chemistry, had been a mistake. Therefore, one will nowadays be less inclined to assume that the concepts of physics, even those of quantum theory, can certainly be applied everywhere in biology or other sciences. We will, on the contrary, try to keep the doors open for the entrance of new concepts even in those parts of science where the older concepts have been very useful for the understanding of the phenomena. Especially at those points where the application of the older concepts seems somewhat forced or

appears not quite adequate to the problem we will try to avoid any rash conclusions.

Furthermore, one of the most important features of the development and the analysis of modern physics is the experience that the concepts of natural language, vaguely defined as they are, seem to be more stable in the expansion of knowledge than the precise terms of scientific language, derived as an idealization from only limited groups of phenomena. This is in fact not surprising since the concepts of natural language are formed by the immediate connection with reality; they represent reality. It is true that they are not very well defined and may therefore also undergo changes in the course of the centuries, just as reality itself did, but they never lose the immediate connection with reality. On the other hand, the scientific concepts are idealizations; they are derived from experience obtained by refined experimental tools, and are precisely defined through axioms and definitions. Only through these precise definitions is it possible to connect the concepts with a mathematical scheme and to derive mathematically the infinite variety of possible phenomena in this field. But through this process of idealization and precise definition the immediate connection with reality is lost. The concepts still correspond very closely to reality in that part of nature which had been the object of the research. But the correspondence may be lost in other parts containing other groups of phenomena.

Keeping in mind the intrinsic stability of the concepts of natural language in the process of scientific development, one sees that—after the experience of modern physics—our attitude toward concepts like mind or the human soul or life or God will be different from that of the nineteenth century, because these concepts belong to the natural language and have therefore

immediate connection with reality. It is true that we will also realize that these concepts are not well defined in the scientific sense and that their application may lead to various contradictions, for the time being we may have to take the concepts, unanalyzed as they are; but still we know that they touch reality. It may be useful in this connection to remember that even in the most precise part of science, in mathematics, we cannot avoid using concepts that involve contradictions. For instance, it is well known that the concept of infinity leads to contradictions that have been analyzed, but it would be practically impossible to construct the main parts of mathematics without this concept.

The general trend of human thinking in the nineteenth century had been toward an increasing confidence in the scientific method and in precise rational terms, and had led to a general skepticism with regard to those concepts of natural language which do not fit into the closed frame of scientific thought—for instance, those of religion. Modern physics has in many ways increased this skepticism; but it has at the same time turned it against the overestimation of precise scientific concepts, against a too-optimistic view on progress in general, and finally against skepticism itself. The skepticism against precise scientific concepts does not mean that there should be a definite limitation for the application of rational thinking. On the contrary, one may say that the human ability to understand may be in a certain sense unlimited. But the existing scientific concepts cover always only a very limited part of reality, and the other part that has not yet been understood is infinite. Whenever we proceed from the known into the unknown we may hope to understand, but we may have to learn at the same time a new meaning of the word “understanding.” We know that any understanding must be based finally upon the natural language because it is only

there that we can be certain to touch reality, and hence we must be skeptical about any skepticism with regard to this natural language and its essential concepts. Therefore, we may use these concepts as they have been used at all times. In this way modern physics has perhaps opened the door to a wider outlook on the relation between the human mind and reality.

This modern science, then, penetrates in our time into other parts of the world where the cultural tradition has been entirely different from the European civilization. There the impact of this new activity in natural and technical science must make itself felt even more strongly than in Europe, since changes in the conditions of life that have taken two or three centuries in Europe will take place there within a few decades. One should expect that in many places this new activity must appear as a decline of the older culture, as a ruthless and barbarian attitude, that upsets the sensitive balance on which all human happiness rests. Such consequences cannot be avoided; they must be taken as one aspect of our time. But even there the openness of modern physics may help to some extent to reconcile the older traditions with the new trends of thought. For instance, the great scientific contribution in theoretical physics that has come from Japan since the last war may be an indication for a certain relationship between philosophical ideas in the tradition of the Far East and the philosophical substance of quantum theory. It may be easier to adapt oneself to the quantum-theoretical concept of reality when one has not gone through the naïve materialistic way of thinking that still prevailed in Europe in the first decades of this century.

Of course such remarks should not be misunderstood as an underestimation of the damage that may be done or has been done to old cultural traditions by the impact of technical prog-

ress. But since this whole development has for a long time passed far beyond any control by human forces, we have to accept it as one of the most essential features of our time and must try to connect it as much as possible with the human values that have been the aim of the older cultural and religious traditions. It may be allowed at this point to quote a story from the Hasidic religion: There was an old rabbi, a priest famous for his wisdom, to whom all people came for advice. A man visited him in despair over all the changes that went on around him, deploring all the harm done by so-called technical progress. "Isn't all this technical nuisance completely worthless," he exclaimed "if one considers the real values of life?" "This may be so," the rabbi replied, "but if one has the right attitude one can learn from everything." "No," the visitor rejoined, "from such foolish things as railway or telephone or telegraph one can learn nothing whatsoever." But the rabbi answered, "You are wrong. From the railway you can learn that you may by being one instant late miss everything. From the telegraph you can learn that every word counts. And from the telephone you can learn that what we say here can be heard there." The visitor understood what the rabbi meant and went away.

Finally, modern science penetrates into those large areas of our present world in which new doctrines were established only a few decades ago as foundations for new and powerful societies. There modern science is confronted both with the content of the doctrines, which go back to European philosophical ideas of the nineteenth century (Hegel and Marx), and with the phenomenon of uncompromising belief. Since modern physics must play a great role in these countries because of its practical applicability, it can scarcely be avoided that the narrowness of the doctrines is felt by those who have really understood modern

physics and its philosophical meaning. Therefore, at this point an interaction between science and the general trend of thought may take place. Of course the influence of science should not be overrated; but it might be that the openness of modern science could make it easier even for larger groups of people to see that the doctrines are possibly not so important for the society as had been assumed before. In this way the influence of modern science may favor an attitude of tolerance and thereby may prove valuable.

On the other hand, the phenomenon of uncompromising belief carries much more weight than some special philosophical notions of the nineteenth century. We cannot close our eyes to the fact that the great majority of the people can scarcely have any well-founded judgment concerning the correctness of certain important general ideas or doctrines. Therefore, the word "belief" can for this majority not mean "perceiving the truth of something" but can only be understood as "taking this as the basis for life." One can easily understand that this second kind of belief is much firmer, is much more fixed than the first one, that it can persist even against immediate contradicting experience and can therefore not be shaken by added scientific knowledge. The history of the past two decades has shown by many examples that this second kind of belief can sometimes be upheld to a point where it seems completely absurd, and that it then ends only with the death of the believer. Science and history can teach us that this kind of belief may become a great danger for those who share it. But such knowledge is of no avail, since one cannot see how it could be avoided, and therefore such belief has always belonged to the great forces in human history. From the scientific tradition of the nineteenth century one would of course be inclined to hope that all belief should be based on a

rational analysis of every argument, on careful deliberation; and that this other kind of belief, in which some real or apparent truth is simply taken as the basis for life, should not exist. It is true that cautious deliberation based on purely rational arguments can save us from many errors and dangers, since it allows readjustment to new situations, and this may be a necessary condition for life. But remembering our experience in modern physics it is easy to see that there must always be a fundamental complementarity between deliberation and decision. In the practical decisions of life it will scarcely ever be possible to go through all the arguments in favor of or against one possible decision, and one will therefore always have to act on insufficient evidence. The decision finally takes place by pushing away all the arguments—both those that have been understood and others that might come up through further deliberation—and by cutting off all further pondering. The decision may be the result of deliberation, but it is at the same time complementary to deliberation; it excludes deliberation. Even the most important decisions in life must always contain this inevitable element of irrationality. The decision itself is necessary, since there must be something to rely upon, some principle to guide our actions. Without such a firm stand our own actions would lose all force. Therefore, it cannot be avoided that some real or apparent truth form the basis of life; and this fact should be acknowledged with regard to those groups of people whose basis is different from our own.

Coming now to a conclusion from all that has been said about modern science, one may perhaps state that modern physics is just one, but a very characteristic, part of a general historical process that tends toward a unification and a widening of our present world. This process would in itself lead to a diminution

of those cultural and political tensions that create the great danger of our time. But it is accompanied by another process which acts in the opposite direction. The fact that great masses of people become conscious of this process of unification leads to an instigation of all forces in the existing cultural communities that try to ensure for their traditional values the largest possible role in the final state of unification. Thereby the tensions increase and the two competing processes are so closely linked with each other that every intensification of the unifying process—for instance, by means of new technical progress—intensifies also the struggle for influence in the final state, and thereby adds to the instability of the transient state. Modern physics plays perhaps only a small role in this dangerous process of unification. But it helps at two very decisive points to guide the development into a calmer kind of evolution. First, it shows that the use of arms in the process would be disastrous and, second, through its openness for all kinds of concepts it raises the hope that in the final state of unification many different cultural traditions may live together and may combine different human endeavors into a new kind of balance between thought and deed, between activity and meditation.